DRL-Driven STAR-RIS Assisted Secure Communication With HPPP Eavesdroppers

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***Abstract*— This letter explores a secure communication system assisted by simultaneously transmitting and reflecting reconfigurable intelligent surfaces (STAR-RIS) with coupled phase-shift design. Here, a multi-antenna base station (BS) transmits confidential information to both indoor and outdoor users via STAR-RIS using an energy-splitting protocol. We consider multiple terrestrial homogeneous Poisson point process eavesdroppers located in reflecting and transmitting regions that can intercept the BS's transmitted signal. To ensure secure communication, we formulate a secrecy rate maximization problem by jointly optimizing the BS beamforming vectors and the transmission and reflection coefficients of the STAR-RIS. Furthermore, we reformulate the optimization problem as a constrained Markov decision process, considering both system dynamics and problem tractability. To solve it efficiently, we employ a deep reinforcement learning technique, i.e., the constrained soft actor-critic algorithm. Numerical analysis confirms that the proposed framework achieves superior secrecy performance compared to benchmark methods.**

***Index Terms*— Reconfigurable intelligent surfaces, coupled phase shift, security, deep reinforcement learning.**

I. Introduction

RECONFIGURABLE intelligent surfaces (RIS)-assisted communications have emerged to meet the stringent requirements of sixth-generation wireless networks, such as ultra-high energy efficiency, increased data rates, ultra-reliable communication, and ultra-low latency [1]. They can be utilized in various real-world applications to improve target sensing, energy transfer, and communication efficiency across diverse environments [2], [3]. However, the functionality of conventional RIS is limited to a single half-space (i.e., 180° coverage), restricting their applicability in scenarios that require omnidirectional coverage [4]. To expand coverage from 180° to 360°, a simultaneous transmitting and reflecting reconfigurable intelligent surface (STAR-RIS) architecture has been introduced that enables full-space signal control by simultaneously transmitting and reflecting incident waves [5].

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On the other hand, the expanded coverage of STAR-RIS introduces security vulnerabilities due to the inherent broadcast nature of wireless communication, which makes it prone to security threats [6]. With STAR-RIS enabling both transmission and reflection, eavesdroppers can exploit signals from both sides of the surface, thereby significantly increasing information leakage [7]. Numerous studies have been conducted to enhance the security of STAR-RIS-assisted communication networks by leveraging physical-layer security techniques [7], [8], [9], [10]. However, these works considered independently adjustable transmission and reflection coefficients (TRCs) that typically require complex semi-passive or active STAR-RIS architectures.

The concept of coupled phase shifts in STAR-RISs has been explored in [11], where a secure beamforming strategy was formulated to address full-space mutual eavesdropping. Furthermore, [12] presented a secure beamforming design specifically designed for these coupled-phase STAR-RIS scenarios, utilizing a penalty-based optimization framework. Conventional and model-based optimization methods, as discussed in [11], [12], often struggle with high-complexity and non-convex problems. In this context, the authors in [13] and [14] implemented deep reinforcement learning (DRL) frameworks to improve the secrecy performance of STAR-RIS-assisted wireless systems. Specifically, the authors in [13] assumed that the eavesdropper locations and channel state information (CSI) were perfectly known. Whereas, the authors in [14] considered the presence of a legitimate user and an eavesdropper in the reflection and transmission region, respectively. However, in practical scenarios, eavesdroppers can be randomly located within the network. Thus, stochastic spatial models such as the homogeneous Poisson point process (HPPP) [15] or similar stochastic-geometry approaches [10] are required to characterize their random locations.

Motivated by the identified research gap, we explore a STAR-RIS-assisted secure communication system with coupled phase shifts, considering spatially random HPPP eavesdroppers located on either side of the STAR-RIS. We consider a multiple-antenna base station (BS) that transmits confidential data to legitimate users located in the reflecting and transmitting regions via a STAR-RIS using the energy splitting (ES) protocol. In the considered model, the users depend entirely on the STAR-RIS link for communication. With this setup, we maximize the sum secrecy rate (SSR) by optimizing the TRCs of STAR-RIS elements and digital beamforming weights at the BS under practical design constraints. Unlike [11], [12], we reformulate the original problem as a

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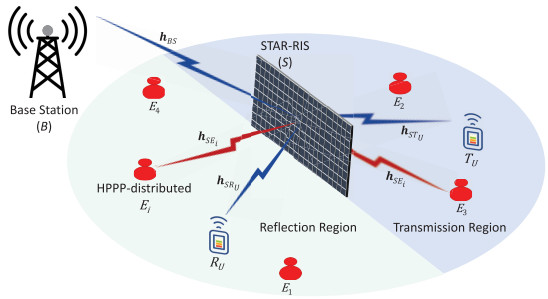

Fig. 1. STAR-RIS assisted secure network.

constrained Markov decision process (CMDP) and solve it using a constrained soft actor-critic (CSAC)-based framework.

II. System Model Discussion

A. System and Channel Model

We consider a STAR-RIS-assisted secure network as illustrated in Fig. 1. The system consists of a STAR-RIS (S), a BS (B), and two terrestrial legitimate users[1] R_U and T_U that are located in the reflection and transmission regions of S, respectively. We adopt the ES protocol for S to simultaneously serve R_U and T_U. Consequently, each element of S splits the incident signal from B into both transmitted and reflected components. Additionally, S consists of K passive elements, and B is equipped with N antennas, while all other nodes are assumed to be equipped with a single antenna. The direct links between $B-R_U$ and $B-T_U$ are severely hindered by obstacles such as buildings and trees. As a result, R_U and T_U rely solely on S to realize communication from B. There are multiple terrestrial eavesdroppers, denoted as E_i, $\forall E_i \in \Phi_e$, randomly distributed within the radial coverage of S with radius R_e, following HPPP with density λ_e [15]. Moreover, E_i can be present in any regions of S, and therefore, we associate a Bernoulli random variable, $b_i \in \{0,1\}$, with each eavesdropper. Specifically, $b_i = 1$ with probability $P_{i,R}$ indicates that the i-th eavesdropper is in the reflection region, while $b_i = 0$ with probability $P_{i,T}$ means it is in the transmission region.

We consider the Rician fading model for the communication links, i.e., $B-S$ and $S-l$, $\forall l \in \{R_U, T_U, E_i\}$ [8], [16]. The channel matrix for the $B-S$ link is denoted as $\mathbf{h}_{B,S} \in \mathbb{C}^{N\times K}$ and the channel coefficients between S and l are denoted as $\mathbf{h}_{S,l} \in \mathbb{C}^{K\times 1}$, can be given as $\mathbf{h}_{p,q} = \sqrt{\rho_0 d_{p,q}^{-\alpha}}(\sqrt{\frac{\mathcal{K}}{1+\mathcal{K}}}\mathbf{g}_{p,q}^{\text{LoS}} + \sqrt{\frac{1}{1+\mathcal{K}}}\mathbf{g}_{p,q}^{\text{NLoS}})$, where $(p,q) \in \{(B,S),(S,l)\}$, ρ_0 is the reference channel gain at a distance of 1 m, $d_{B,S}$ and $d_{S,l}$ denote distances, $\mathcal{K}$ represents the Rician factor. $\mathbf{g}_{B,S}^{\text{LoS}}$ and $\mathbf{g}_{S,l}^{\text{LoS}}$ are the LoS components and $\mathbf{g}_{B,S}^{\text{NLoS}} \sim \mathcal{CN}(0,1)$, $\mathbf{g}_{S,l}^{\text{NLoS}} \sim \mathcal{CN}(0,1)$ are the NLoS components [13], [16], [17].

Now, we define the transmission and reflection coefficients (TRCs) matrices of S, which are expressed as

$$\mathbf{\Phi}^{\xi}(t) = \text{diag}\left(\sqrt{\beta_1^{\xi}(t)}e^{j\phi_1^{\xi}}(t), \ldots, \sqrt{\beta_N^{\xi}(t)}e^{j\phi_K^{\xi}}(t)\right), \quad (1)$$

where $\xi \in \{T,R\}$. The corresponding TRCs are denoted as $\beta_k^T, \beta_k^R \in [0,1]$ denote the amplitude coefficients, whereas $\phi_k^T, \phi_k^R \in [0,2\pi)$ represent the phase shifts associated with the k-th element. Additionally, we adopt a passive and lossless hardware structure for S as in [7] and [10]. Under this model, the coupled behavior of the TRCs satisfies $(\beta_k^T)^2 + (\beta_k^R)^2 = 1$. This relationship is derived from the law of energy conservation and the condition for the coupled phase shift is expressed as $|\phi_k^R - \phi_k^T| = \frac{\pi}{2}$ or $\frac{3\pi}{2}$ [12], [13]. We assume that the channel state information of different nodes can be obtained using techniques such as RIS-based pilot training [7], [10], [13] or compressed sensing-based estimation [14].

[1]The proposed scheme can be extended to a generalized scenario involving multiple indoor and outdoor users and eavesdroppers [13].

B. Data Transmission Model

A multi-antenna BS transmits confidential information $x_\zeta(t), \forall \zeta \in \{R_U, T_U\}$ during the t-th time slot, using the beamforming vector $w_\zeta(t) \in \mathbb{C}^{N\times 1}$ with total transmit power as $\sum_\zeta \|w_\zeta(t)\|^2$. Using the ES protocol, each element of S reflects and transmits portion of the incident signal toward R_U and T_U, respectively. Therefore, the composite channel from B to user ζ is denoted as $\mathbf{h}_{B,\zeta}(t) = (\mathbf{h}_{S,\zeta})^H(t)\mathbf{\Phi}^{\xi}(t)\mathbf{h}_{B,S}(t)$, where $\xi = R$ for $\zeta = R_U$ and $\xi = T$ for $\zeta = T_U$. So, the received signal at user ζ is expressed as $y_\zeta(t) = \mathbf{h}_{B,\zeta}(t)\sum_\zeta w_\zeta(t)x_\zeta(t) + n_\zeta(t)$, where $n_\zeta(t) \sim \mathcal{CN}(0,\sigma_b^2)$ denotes additive white Gaussian noise (AWGN) at user ζ. The achievable rate at user ζ during t-th slot is expressed as

$$R_\zeta(t) = \mathbb{E}\left[\log_2\left(1 + \frac{\mathbf{h}_{B,\zeta}(t)w_\zeta(t)\mathbf{h}_{B,\zeta}^H(t)}{\mathbf{h}_{B,\zeta}(t)w_{\bar{\zeta}}(t)\mathbf{h}_{B,\zeta}^H(t) + \sigma_\zeta^2}\right)\right], \quad (2)$$

where $\zeta, \bar{\zeta} \in \{R_U, T_U\}$ with $\bar{\zeta} \neq \zeta$. Due to the broadcast nature, the information intended for R_U and T_U can be leaked to nearby eavesdroppers. The received signal at E_i is given as

$$y_{E_i}(t) = \mathbf{h}_{S,E_i}^H \mathbf{B}\, \mathbf{h}_{B,S}(t)\sum_\zeta w_\zeta(t)x_\zeta(t) + n_{E_i}(t), \quad (3)$$

where $\mathbf{B} = b_i\mathbf{\Phi}^R(t) + (1-b_i)\mathbf{\Phi}^T(t)$, $n_{E_i}(t) \sim \mathcal{CN}(0,\sigma_e^2)$ denotes the AWGN at eavesdropper E_i. Moreover, we define the effective composite channel from B to E_i as $\mathbf{h}_{B,E_i}(t) = (\mathbf{h}_{S,E_i}(t))^H\mathbf{\Phi}^{\xi}(t)\mathbf{h}_{B,S}(t)$, where $\xi = R$ if $b_i = 1$ and $\xi = T$, otherwise. Assuming each E_i is capable of extracting the confidential information for user ζ while treating the other user's signal as interference, the corresponding wiretap rate is expressed as

$$R_{E_i,\zeta}(t) = \mathbb{E}\left[\log_2\left(1 + \frac{\mathbf{h}_{B,E_i}(t)w_\zeta(t)\mathbf{h}_{B,E_i}^H(t)}{\mathbf{h}_{B,E_i}(t)w_{\bar{\zeta}}(t)\mathbf{h}_{B,E_i}^H(t) + \sigma_e^2}\right)\right]. \quad (4)$$

We analyze the worst-case eavesdropping scenario by considering the highest rate among all eavesdroppers, denoted as $R_{e,\zeta}^{\max}(t) = \max_{E_i\in\Phi_e} R_{E_i,\zeta}(t)$. Accordingly, the instantaneous secrecy rate for user ζ is given as $R_{s,\zeta}(t) = \left[R_\zeta(t) - R_{e_i,\zeta}^{\max}(t)\right]^+$, where $[x]^+ = \max(x,0)$. The SSR for the system in t-th slot is calculated by adding the individual secrecy rates of both users, given as $R_s^{\text{sum}}(t) = R_s^{R_U}(t) + R_s^{T_U}(t)$.

C. Problem Formulation

We aim to maximize the SSR of the considered model, which is achieved by jointly optimizing the beamforming

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vectors at the BS ($w_{R_U}(t)$ and $w_{T_U}(t)$), and TRC matrices of the S ($\mathbf{\Phi}^T(t)$) and $\mathbf{\Phi}^R(t)$, respectively. This is subject to the BS's power budget and the physical constraints of the passive, lossless architecture of S. Thus, the SSR maximization problem is formulated as

$$\text{(P1)}: \max_{\{w_\varsigma, \mathbf{\Phi}^\xi\}} \sum_{t=1}^{T} R_s^{\text{sum}}(t), \tag{5a}$$

$$\text{s.t. } (\beta_k^T(t))^2 + (\beta_k^R(t))^2 = 1, \quad \forall k, \forall t, \tag{5b}$$

$$0 \le \beta_k^T(t), \beta_k^R(t) < 1, \quad \forall k, \forall t, \tag{5c}$$

$$|\phi_k^R(t) - \phi_k^T(t)| \in \left\{\frac{\pi}{2}, \frac{3\pi}{2}\right\}, \quad \forall k, \forall t, \tag{5d}$$

$$0 \le \phi_k^T(t), \phi_k^R(t) < 2\pi, \quad \forall k, \forall t, \tag{5e}$$

$$R_\varsigma(t) - R_{e_i,\varsigma}^{\max}(t) \ge 0, \forall t, \tag{5f}$$

$$\sum_{\varsigma} \|\mathbf{w}_\varsigma(t)\|^2 \le P^{\max}, \quad \forall t. \tag{5g}$$

The constraints are as follows: (5b) ensures energy conservation for the elements of S, while (5c) limits transmission and reflection amplitudes. (5d) enforces the phase-shift condition in passive designs of S and (5e) restricts phase shifts. Whereas (5f) guarantees a non-negative instantaneous secrecy rate for each user at all times. (5g) ensures the total transmit power of the BS does not exceed the maximum allowable power $P^{\max}$. We observe that (P1) is inherently non-convex due to the coupled beamforming vectors, the S coefficients, and the non-linear constraints. Additionally, uncertainty in the network arises from the random spatial distribution of eavesdroppers and from channel randomness due to the NLoS components $\mathbf{g}_{B,S}^{\text{NLoS}}$ and $\mathbf{g}_{S,l}^{\text{NLoS}}$. To effectively manage this spatial-temporal randomness, we reformulate (P1) as a CMDP, enabling sequential decision-making while satisfying system constraints.

III. Proposed SAC-Based SSR Maximization Framework

We reformulate the SSR maximization problem as a CMDP to effectively solve under the dynamic system and channel uncertainties, defined by $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \{d_\nu\}_{\nu=1}^m, \{\delta_\nu\}_{\nu=1}^m)$. Specifically, state space ($\mathcal{S}$) at t-slot is given as $s(t) = \{\mathbf{h}_{B,S}(t), \mathbf{h}_{S,R_U}(t), \mathbf{h}_{S,T_U}(t), \mathbf{h}_{S,E_i}(t), \Phi_e, b_i\}$. Followed by, action space ($\mathcal{A}$) includes all optimization variables to be controlled at at t-slot, represented as $a(t) = \{w_{R_U}(t), w_{T_U}(t), \mathbf{\Phi}^T(t), \mathbf{\Phi}^R(t)\}$. The transition probability ($\mathcal{P}$), which governs the system dynamics and is defined as $\mathcal{P}(s(t+1) \mid s(t), a(t))$, also captures the Rician fading randomness, HPPP eavesdroppers, and Bernoulli variables. The reward function ($\mathcal{R}$) at t-slot is given as $R_{s,R_U}(t) + R_{s,T_U}(t)$ if no constraints is violated and -50 otherwise. Moreover, to enforce the system constraints, we define a set of constraint cost functions $d_\nu(s,a)_{\nu=1}^m$ along with their corresponding thresholds δ_ν. Specifically, $d_1(s,a) = \sum_\varsigma \|\mathbf{w}_\varsigma(t)\|^2$, $d_2(s,a) = |(\beta_k^T(t))^2 + (\beta_k^R(t))^2 - 1|$, $d_3(s,a) = |\phi_k^R(t) - \phi_k^T(t)| - \frac{\pi}{2}$ or $\frac{3\pi}{2}$, and $d_4(s,a) = \max(0, R_{e_i,\varsigma}(t) - R_\varsigma(t))$ with $\delta_1 = P^{\max}$, $\delta_2 = 0$, with $\delta_3 = 0$, and with $\delta_4 = 0$. The action space incorporates continuous-valued beamforming, along with discrete and coupled TRCs, under various constraints; therefore, we employ a DRL algorithm to solve the CMDP.

A. Proposed CSAC-Based Learning Framework

To effectively solve the SSR maximization problem formulated as a CMDP, we adopt a CSAC algorithm. This algorithm combines the entropy-regularized structure of the SAC with a Lagrangian relaxation mechanism, allowing for constraint satisfaction in uncertain environments [13]. The standard SAC algorithm can not handle the CMDP, which comprises an instantaneous costs' vector $\mathbf{C}(t) = [C_1(t), \ldots, C_7(t)]^\mathsf{T}$, each associated with a constraint threshold d_κ and a Lagrange multiplier λ_κ. To handle these, the CSAC transforms the constrained optimization into an unconstrained saddle-point problem given as

$$\min_{\boldsymbol{\lambda} \ge 0} \max_{\pi_\chi} \mathbb{E}_{\pi_\chi}\left[\sum_{t=0}^{T-1} \gamma^t \Big(R(s[t], a[t]) - \sum_{\kappa=1}^{7} \lambda_\kappa C_\kappa(s[t], a[t]) + \alpha \mathcal{H}\left(\pi_\chi(\cdot|s[t])\right) \Big)\right], \tag{6}$$

where $\pi_\chi(a|s)$ is the stochastic policy parameterized by neural network weights (χ), entropy term $\mathcal{H}(\pi_\chi(\cdot|s[t]))$ encourages policy randomness, $\gamma \in (0,1)$ is the discount factor, and $\alpha > 0$ is the entropy temperature controlling the degree of exploration. To estimate the long-term return, we utilize a single reward critic, denoted as $Q_{R,\theta}(s,a)$. Additionally, we have seven cost critics, represented as $Q_{C_\kappa,\theta_\kappa}(s,a)$, to monitor cumulative constraint violations. These networks are trained using target Q-learning, employing Bellman updates and Polyak averaging. The policy parameters are updated by maximizing the Lagrangian-augmented objective, obtained as

$$\mathcal{J}_\pi = \mathbb{E}_{(s,a)\sim\mathcal{B}}\left[\alpha \log \pi_\chi(a|s) - Q_{R,\theta}(s,a) + \sum_{\kappa=1}^{7} \lambda_\kappa Q_{C_\kappa,\theta_\kappa}(s,a)\right], \tag{7}$$

where $\mathcal{B}$ is a mini-batch sampled from the replay buffer $\mathcal{D}$. The dual multipliers are updated via projected gradient ascent to enforce the long-term cost thresholds:

$$\lambda_\kappa^{(t+1)} = \left[\lambda_\kappa^{(t)} + \Upsilon_\lambda \left(\frac{1}{|\mathcal{B}|} \sum_{(s,a,\mathbf{C})\in\mathcal{B}} C_\kappa - d_\kappa\right)\right]^+, \tag{8}$$

where Υ_λ is the dual learning rate and $[\cdot]^+$ denotes the ReLU projection to ensure $\lambda_\kappa \ge 0$. The CSAC learning procedure is detailed in Algorithm 1. It alternates between fast updates of $(\chi, \theta, \theta_\kappa)$ and slower updates of $\boldsymbol{\lambda}$, leading to convergence towards a local saddle point of the Lagrangian. During offline training, each gradient update step involves one forward-and-backward pass through the policy network, one for the reward ($Q_{R,\theta}$) and seven for the constraints ($Q_{C_\kappa,\theta_\kappa}$, for $\kappa = 1, \ldots, 7$). Moreover, the spatial randomness of HPPP eavesdroppers is implicitly learned by the CSAC agent through repeated exposure to varying spatial realizations. This enables adaptive policy learning that remains robust to spatial uncertainty in real time.

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Algorithm 1 Proposed CSAC-Based Framework

Initialize policy network parameters, $E_m = 10000$.
for episode $= 1$ to E_m **do**
- Initialize environment and observe initial state $s(0)$.
- **for** time slot t in episode **do**
  1) Sample action $a(t) \sim \pi_\chi(\cdot|s(t))$.
  2) Execute $a(t)$, receive reward $R(t)$, costs $\mathbf{C}(t) = [C_1(t), \ldots, C_7(t)]^\mathsf{T}$, and observe next state $s(t+1)$.
  3) Store transition $(s(t), a(t), R(t), \mathbf{C}(t), s(t+1))$ in replay buffer $\mathcal{D}$.
  4) Sample mini-batch $\mathcal{B}$ from $\mathcal{D}$.
  5) Update reward critic $Q_{R,\theta}$ and $Q_{C_\kappa,\theta_\kappa}$ using target Q-learning and Polyak averaging.
  6) Update policy parameters χ by maximizing the objective in (7).
  7) Compute mini-batch cost averages and update dual variables via (8).
  8) **if** $s(t+1)$ is terminal, **end**.

end

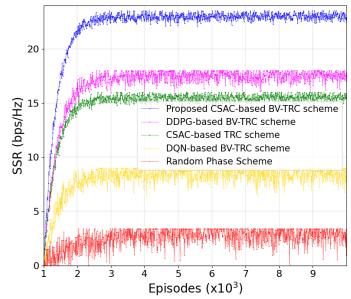

Fig. 2. Convergence nature of proposed and baseline schemes.

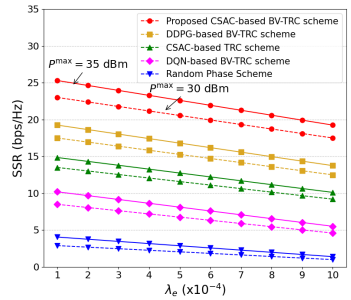

Fig. 3. SSR versus λ_e curves for different schemes.

B. Computational Complexity

Let each network consist of J fully connected layers with widths $H_j{}_{j=1}^{J}$. Therefore, the computational cost of one pass through a single network is $\mathcal{O}\left(\sum_{j=1}^{J-1} H_j H_{j+1}\right)$. Assuming a mini-batch size of $\mathcal{B}$ and N_{up} total stochastic gradient updates during training, the overall training-time complexity becomes $\mathcal{O}(\mathcal{B} N_{\text{up}} \sum_{j=1}^{J-1} H_j H_{j+1})$. During the online execution phase, the agent infers actions based on the current policy and evaluates the critic networks. In the t-th time slot, the agent generates beamforming vectors and TRCs through a forward pass in the actor network. The cost of a single forward pass through the actor is $\mathcal{O}(C_\pi) = \mathcal{O}\left(\sum_{j=1}^{J-1} H_j H_{j+1}\right)$. In the online decision-making process, the cost per t-th slot remains directly proportional to the size of the network. While CSAC incurs a higher offline training cost ($\mathcal{O}(9\mathcal{B} N_{\text{up}} C_{\text{net}})$) due to multiple critics, its online inference complexity remains low, comparable to DDPG and more efficient than DQN, thereby achieving superior adaptability and overall performance.

IV. Simulation Results and Discussions

In this section, we demonstrate the effectiveness of our proposed algorithm through simulation results, using setups similar to those in [13], [14], and [17]. The neural network architecture has two fully connected layers with 256 hidden units per layer, using the ReLU activation function and Adam optimizer. The simulation hyperparameters are: $\gamma = 0.95$, $E_m = 10000$, $\mathcal{D} = 10^5$, $\mathcal{B} = 64$, learning rates for the policy network, critic network, and Lagrangian multipliers are set to 10^{-3}, 10^{-3}, and 10^{-4}, respectively. The distance between the involved nodes are given as: $d_{B,S} = 200$ m, $d_{S,T_U} = 75$ m, and $d_{S,R_U} = 100$ m, whereas for the HPPP eavesdroppers, $R_e = 200$ m and $\lambda_e = 10^{-4}$. The channel gain at the reference distance is set as $\rho_0 = -20$ dBm, $\mathcal{K} = 10$, $\alpha = 2.6$, and $P^{\max} = 30$ dBm. The proposed algorithm maximizes the system's SSR by jointly optimizing the beamforming vectors at the BS and TRC matrices of S using the CSAC algorithm (*CSAC-based BV-TRC scheme*). To show the effectiveness of the proposed framework, we consider the following benchmark schemes. Specifically, for *CSAC-based TRC scheme*, we optimize TRC matrices using the CSAC algorithm without beamforming vectors. Then, *DQN-based BV-TRC scheme* optimizes beamforming vectors and TRC matrices using the DQN algorithm. Next, *DDPG-based BV-TRC scheme* optimizes beamforming vectors and TRC matrices using the DDPG algorithm. Lastly, *Random phase scheme* optimizes beamforming vectors with randomly-assigned phase shifts of S.

Fig. 2 demonstrates the convergence behavior of the proposed and baseline schemes in terms of SSR over $10,000$ training episodes. The proposed *CSAC-based BV-TRC scheme* quickly converges to a high SSR value, confirming the learning efficiency and robustness of the SAC algorithm. In contrast, *DDPG-based BV-TRC scheme* attains comparatively lower SSR value. The *CSAC-based TRC scheme*, which optimizes only TRCs without considering beamforming. The *DQN-based* and *Random Phase* schemes achieve even lower SSRs because they struggle to explore high-dimensional continuous action spaces effectively.

Fig. 3 evaluates the impact of increasing λ_e on the SSR for different schemes under two power budgets, i.e., 30 dBm and 35 dBm. As λ_e increases, the SSR decreases across all schemes because a higher density of randomly distributed HPPP eavesdroppers increases the likelihood of interception and overall secrecy degradation. However, the proposed *CSAC-based BV-TRC scheme* demonstrates greater robustness against spatial randomness owing to its adaptive policy-learning mechanism, which enables the agent to optimize beamforming and transmission reflection coefficients across diverse eavesdropper realizations. Moreover, the curves show that the SSR values are higher for $P^{\max} = 35$ dBm than for

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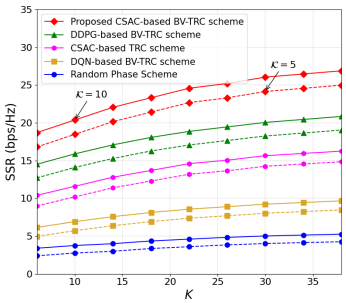

Fig. 4. SSR versus K curves for different $\mathcal{K}$.

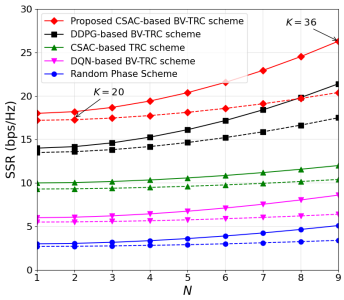

Fig. 5. SSR versus N curves for different STAR-RIS elements.

$P^{\max} = 30$ dBm across all schemes, as a higher power budget allows the BS to allocate more power to beamforming weights, thereby improving the received signal strength at legitimate users.

In Fig. 4, we examine the impact of the number of STAR-RIS elements (K) on the SSR performance for $\mathcal{K} = 5$ and $\mathcal{K} = 10$. The curves show a rising trend for all schemes as K increases. The proposed *CSAC-based BV-TRC scheme* achieves the highest SSR in both scenarios, as increasing K boosts the passive beamforming capability of S. With $\mathcal{K} = 10$, a stronger line-of-sight component enhances channel gains between legitimate nodes. Consequently, all schemes perform better with $\mathcal{K} = 10$ than with $\mathcal{K} = 5$.

Lastly, Fig. 5 illustrates the impact of increasing the number of antennas at B on the SSR performance for two different configurations of S, i.e., $K = 20$ and $K = 36$. As N increases, the SSR performance improves for all schemes due to the enhanced active beamforming capabilities of B. Specifically, the proposed scheme shows the most significant improvement in the SSR value, achieving SSR of 26.5 bps/Hz at $K = 36$ and 20.4 bps/Hz at $K = 20$. This improvement is due to the joint optimization of STAR-RIS coefficients and beamforming at B. Whereas, *CSAC-based TRC scheme* attains suboptimal SSR values due to overlooking of BS active beamforming. The *DQN-based BV-TRC scheme* struggles with poor convergence and limited exploration. Lastly, the *random phase scheme* performs the worst due to a lack of intelligent control.

V. Conclusion

This work investigated a STAR-RIS-assisted secure wireless communication system with a coupled phase-shift architecture. By leveraging an energy-splitting protocol, we enabled a multi-antenna BS to transmit confidential information simultaneously to indoor and outdoor users. We considered randomly distributed HPPP eavesdroppers located across transmission and reflection zones. We formulated an optimization problem to maximize the sum secrecy rate by optimizing the BS beamforming vectors and the STAR-RIS transmission and reflection coefficients. The problem was reformulated as a CMDP, and a SAC-based framework is developed to learn optimal policies efficiently. Simulation results demonstrated that the proposed CSAC-based framework significantly outperforms baselines in terms of secrecy performance.

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